# SEÁN Ó MURCHÚ

## Senior Software Engineer

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Backend engineer with eight years of experience designing distributed, event-driven systems. Specialises in functional programming, observability, and developer experience.

# EXPERIENCE

### Senior Software Engineer

**Acme Corp**

Jun 2022 – Present  Dublin, Ireland

*Platform team lead. Owns the event-sourcing stack underpinning the wider product surface.*

- Led the migration of a customer-facing monolith into a set of event-driven microservices, halving p99 latency.
- Designed and rolled out an event-sourcing platform now used by four product teams.
- Mentored three engineers from mid to senior level.

Scala | Go | Kafka | Kubernetes

### Software Engineer

**Liffey Labs**

Jun 2019 – Jun 2021  Remote

- Built and shipped the first version of a SaaS product from scratch alongside a two-person team.
- Introduced the CI/CD pipeline and developer-onboarding flow that scaled the engineering org from 3 to 15.

Python | Go | AWS | GitHub Actions

# VOLUNTEER

### Mentor

**CoderDojo Dublin**

Jun 2020 – Present

*Volunteer-led coding club introducing young people to programming.*

- Weekly mentoring sessions for 10–14 year-olds learning to code.
- Curate the Scala / functional-programming track.

# PROJECTS

### Application

*Hyperion*

Acme Corp

*Distributed task scheduler in Rust*

**Lead** | **Architect**

Jan 2024 – Aug 2024

- Handled 10k tasks/s on a single node.
- Designed an idempotent retry protocol.

Rust | Tokio | gRPC

# AREAS OF FOCUS

- Distributed systems: Kafka, CDC, idempotent consumers, event sourcing.
- Functional programming and type-driven design in Scala and Haskell.
- Developer experience: tooling, CI/CD, observability.
- Engineering culture: ADRs, RFCs, mentoring.

# SKILLS

**Languages** - Scala | Haskell | Python | Go | Rust

**Infra** - Kafka | AWS | Terraform | Docker | Kubernetes

# LANGUAGES

English

Irish

French

Portuguese

# EDUCATION

### M.Sc. in Computer Science

**Tallaght Institute of Technology**

2017 – 2019

Distinction | Distributed Systems | Type Theory | Concurrency

### Computer Science

**Trinity College Dublin**

2014 – 2017

### *Crucible*

*Migration tool for legacy databases*

Maintainer

Sep 2024 – Present

- Schema diffing across PostgreSQL and MySQL.

**Library**

### *Tinkerbell*

*Tiny IRC bot*

Lua

**Other**

### *Quill*

## PUBLICATIONS

### Articles

- *Event Sourcing in Practice*.
  Personal Blog
  14 Jun 2024
  A walk-through of idempotent Kafka consumers and the replay strategy that ships with them.
- *Untyped Note*.
  Jan 2024

### Conference Papers

- *Idempotent Kafka Consumers at Scale*.
  EuroSys
  Apr 2024

### Talks

- *Designing for Failure*.
  ScalaIO
  Sep 2023

### Books

- *Functional Domain Modelling*.
  Self-published
  2022

## INTERESTS

Music - Trad Jazz

Sport - Hurling Climbing

Travel - Japan Iceland

## CERTIFICATIONS

CNCF

Certified Kubernetes Administrator · Aug 2023

Certified Kubernetes Application Developer · Jan 2024

AWS

AWS Solutions Architect — Professional · May 2023

AWS DevOps Engineer — Professional · Sep 2024

Hashicorp Terraform Associate · Dec 2022

Certified Kubernetes Security Specialist · Mar 2024

Google Cloud Professional Architect · Nov 2023

## AWARDS

### *Best Paper Award*

**ScalaConf**

Sep 2024

For *Idempotent Kafka Consumers*.

### *Dean's List*

## REFERENCES

Dr Ada Lovelace

*Seán is a calm, methodical engineer who left every system he touched in a better state than he found it. Recommended without reservation.*

*Pragmatic, curious, unfailingly kind — would hire again in a heartbeat.*